THE GALOIS ACTION ON THE SEVEN M_{23} HURWITZ POINTS

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Abstract. Huang, Jackson, Lee, Poonen, Pries, and Zhang recently realized M_{23} as a Galois group over $\mathbb{Q}$ by finding a $G_{\mathbb{Q}}$-fixed point among the seven inner Nielsen classes of type $(2A, 23A, 23B)$, and noted that they had no conceptual explanation for this fixed point. We determine the Galois action on the full seven-point Hurwitz fiber. The odd cardinality of this $G_{\mathbb{Q}}$-stable fiber already forces a real Hurwitz point, since complex conjugation is an involution. We compute its action exactly. With branch points $(0, \sqrt{-23}, -\sqrt{-23})$, it acts on the seven classes as $(1\,2)(5\,7)$, fixing 3, 4, 6. Thus there are exactly three real Hurwitz points; the existence of a $\mathbb{Q}$-point is a further arithmetic fact. We then compute the associated genus-4 quotient curves by numerical uniformization and construct marked Petri invariants which we prove algebraic and G_K-equivariant for $K = \mathbb{Q}(\sqrt{-23})$. Trace and norm along the marking descend these to algebraic $G_{\mathbb{Q}}$-equivariant invariants of the unmarked fiber. Integer-relation recognition, independently replicated by two descended coordinates with ~ 70 digits of confirmation margin, gives a high-precision numerical determination of the arithmetic decomposition

$$\{1, \dots, 7\} = \{6\} \sqcup \{1, 2, 3, 4, 5, 7\}.$$

The two coordinates generate the same sextic residue field F; it has signature $(2, 2)$, and its Galois closure has group S_6. Thus, subject only to the stated integer-relation recognitions,

$$\rho(G_{\mathbb{Q}}) = \mathrm{Stab}_{S_7}(6) \cong S_6, \qquad \rho(G_K) = S_6.$$

Arithmetic fixes exactly one point and acts maximally on the remaining six. This symmetry breaking is not forced by the finite-level Grothendieck–Teichmüller constraints: using Guillot's description of $\mathcal{GT}_1(G)$ for finite simple groups, we prove that $\mathcal{GT}_1(M_{23})$ induces the full S_7 on these seven Nielsen classes. Finally, under the same recognition hypothesis, at the prime 23 the seven-point Hurwitz algebra decomposes locally as $\mathbb{Q}_{23} \times F_2 \times F_4$, where F_2 and F_4 are totally ramified tame extensions of degrees 2 and 4. Hence inertia has cycle type $(1)(2)(4)$, making label 6 the unique inertia-fixed point at 23 and giving a concrete local target for a conceptual explanation of the exceptional rational point.

1. Introduction

The inverse Galois problem for the sporadic simple group M_{23} resisted solution longer than for any other sporadic group. It was settled in 2026 by Huang, Jackson, Lee, Poonen, Pries, and Zhang [HJLPPZ], who studied the Hurwitz space of Galois covers of $\mathbb{P}^1$ with monodromy group M_{23}, ramification type $(2A, 23A, 23B)$, and branch points $(0, \sqrt{-23}, -\sqrt{-23})$. The associated inner Nielsen class

$$\mathrm{Ni}(2A, 23A, 23B)^{\mathrm{in}} \;=\; \big\{(x, y, z) \in 2A \times 23A \times 23B : xyz = 1,\ \langle x, y\rangle = M_{23}\big\}/M_{23}$$

consists of exactly seven simultaneous-conjugacy classes; the Hurwitz space is accordingly a degree-7 cover of the configuration base, its fiber above $(0, \sqrt{-23}, -\sqrt{-23})$ consisting of seven points permuted by $G_{\mathbb{Q}} = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. The authors of [HJLPPZ] located a $G_{\mathbb{Q}}$-fixed point among the seven — yielding a regular realization of M_{23} over $\mathbb{Q}$ — and remarked that they "do not have a conceptual explanation" for its existence.

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This note contributes two computations toward that explanation.

Complex conjugation. Before computing the permutation, parity already gives a real point (Corollary 3.1): an involution on seven points can pair off only an even number of them. This forces at least one real embedding of a field of moduli, but neither the number of real points nor a $G_{\mathbb{Q}}$-fixed point.

First, we determine unconditionally how the subgroup generated by complex conjugation acts on the seven points (Theorem 3.2): in our labeling (Convention 2.1) it is the involution

$$\kappa \;=\; (1\,2)(5\,7),$$

fixing the three points $3, 4, 6$. In particular exactly three of the seven points are fixed by the chosen complex conjugation (so their residue fields admit the corresponding real embedding), and the $G_{\mathbb{Q}}$-orbit partition of the seven refines to one of the eight shapes listed in Corollary 3.4. The proof combines the branch cycle argument of Fried [Fri77, DD97] with a finite certificate: after anchoring the $2A$-coordinate, complex conjugation acts on the anchored slice by $y \mapsto x_0 y$, and its effect on the seven classes is witnessed by seven explicit conjugating elements (§3). The three assertions are distinct: real existence is forced by parity, the exact count of three is computed, and rationality requires a point fixed by all of $G_{\mathbb{Q}}$. The rational point of [HJLPPZ] supplies this additional arithmetic input.

The orbit partition. Second, we compute the seven covers themselves. Each quotient curve X_i (the quotient of the Galois cover by a point stabilizer) is a curve of genus 4; we compute its canonical differentials and Petri quadric by the power-series method of Klug–Musty–Schiavone–Voight [KMSV14] at 100 decimal digits, using the standalone Arnoldi solver of [Bel] (§4). From the Petri quadric of the canonical curve we extract *marked invariants*: ratios of quadric coefficients that are invariant both under the projective scale and under the one remaining analytic normalization freedom, hence are algebraic numbers attached to the marked cover, compatible with the Galois action (§5). For the $G_{\mathbb{Q}}$-fixed point (label 6) all six invariants are recognized as elements of $K = \mathbb{Q}(\sqrt{-23})$ with small height — for example

$$J_1 \;=\; \frac{-4152 + 693\sqrt{-23}}{10237},$$

with ~ 85 digits of confirmation margin — validating the method against known ground truth. Applying the same procedure to all seven points and factoring the resulting degree-7 polynomials over K yields the G_K-orbit partition, which combined with κ determines the $G_{\mathbb{Q}}$-partition: it is $1 + 6$, and the permutation image is the maximal group compatible with one fixed point (Numerical Result 6.1).

Verification. Every computational claim in this note is backed by a machine-readable receipt (§8); the census and the κ-action are verified by three independent implementations, including a Magma referee run; a Lean 4 formalization of the finite combinatorial core (kernel-checked `decide` certificates with a single quarantined `native_decide` enumeration) is in progress and is not claimed as part of the verification here. The analytic input — that complex conjugation acts through the branch cycle argument — enters only through the published results cited above.

Relation to [HJLPPZ]. The authors of [HJLPPZ] have announced a revision containing more details, including the action of complex conjugation, which they computed independently. The present computations were performed without access to that revision; the labeled representatives in Appendix A make the two computations directly comparable.

2. The seven Nielsen classes

Convention 2.1. M_{23} denotes the copy $\langle g_1, g_2\rangle \subset S_{23}$ generated by the two permutations of [HJLPPZ], reproduced in Appendix A; $g_1 \in 2A$, $g_2 \in 23A$, $g_3 := (g_1 g_2)^{-1} \in 23B$. Products act on the right of the first factor: $(pq)(i) = p(q(i))$. We write $C = C_{M_{23}}(g_1)$, of order 2688. We use our *own* labels $1, \dots, 7$ for the seven Nielsen classes, fixed by the anchored representatives of Appendix A and chosen so that label 6 is the class of the generating triple printed in [HJLPPZ] (which does not publish a numbering); the appendix permits direct comparison with any numbering in their announced revision.

Proposition 2.2 (census)**.** *The set*

$$T = \big\{(x, y) \in 2A \times 23A : \ (xy)^{-1} \in 23B, \ \langle x, y\rangle = M_{23}\big\}$$

carries exactly seven orbits under simultaneous conjugation, with representatives (g_1, y_1), $\dots$, (g_1, y_7) *listed in Appendix A (numbered as in Convention 2.1). Moreover:*

1. *every orbit meets the anchored slice* $\{(g_1, y)\}$*, and two anchored pairs are conjugate iff they are* C*-conjugate;*
2. *the set* $Y = \{y : (g_1, y) \in T\}$ *has exactly* $18816 = 7 \cdot 2688$ *elements, partitioned into seven* regular C*-orbits (equivalently,* $Z(M_{23}) = 1$ *and each pair has trivial simultaneous centralizer);*
3. *the generation condition is automatic: every* $y \in 23A$ *with* $(g_1 y)^{-1} \in 23B$ *generates* M_{23} *together with* g_1*.*

Proof (certificate). (1) is the orbit–stabilizer transport for the conjugation action on the $2A$-coordinate. (2) and (3) are finite assertions verified by exhaustive enumeration: the classes $23A$, $23B$ (each of size 443 520) are materialized as conjugation orbits, Y is filtered, and the seven C-orbits of the y_i are checked to be disjoint, of size 2688 each, and to cover Y; generation for the seven representatives is certified by verified stabilizer chains (order 10 200 960 for each $\langle g_1, y_i\rangle$), and generation is constant on orbits. The enumeration has been performed by three independent implementations (§8); the in-progress Lean formalization packages the same certificates. □

Remark 2.3. The identity $|Y| = 7 \cdot |C|$, i.e. the regularity of all seven orbits, is equivalent to the vanishing of the simultaneous centralizer of each generating pair, which holds since $Z(M_{23}) = 1$. This is what makes the anchored certificate small: seven orbit representatives, two generators of C, and seven conjugators suffice.

3. Complex conjugation: the involution κ

Write H for the unmarked degree-7 Hurwitz fiber over $\mathbb{Q}$ from the Introduction. Its real points are precisely the geometric points fixed by the chosen complex conjugation. This is the $G_{\mathbb{Q}}$-stable unmarked fiber; the $\sqrt{-23}$-marked data used later are naturally over $K = \mathbb{Q}(\sqrt{-23})$.

Fix the branch points $(b_1, b_2, b_3) = (0, \sqrt{-23}, -\sqrt{-23})$ and a real base point, and order the standard generators of $\pi_1(\mathbb{P}^1 \smallsetminus \{b_1, b_2, b_3\})$ so that the slots carry the classes $(2A, 23A, 23B)$ as in [HJLPPZ]. Complex conjugation fixes b_1 and exchanges $b_2 \leftrightarrow b_3$; the branch cycle argument [Fri77, DD97] computes its effect on Nielsen classes for a conjugation-symmetric bouquet:

$$\kappa(x, y, z) \ = \ (x, \ z^{-1}, \ y^{-1}).$$

(For a triple with $xyz = 1$ this is again a triple with product 1 and classes $(2A, 23A, 23B)$, since inversion exchanges $23A$ and $23B$.[1])

Reversal–inversion gives (z^{-1}, y^{-1}, x^{-1}), in class order $(23A, 23B, 2A)$. Cyclically rotating that tuple to restore the $2A$ slot gives the displayed formula. The anchored certificates below use the restored class order $(2A, 23A, 23B)$.

Corollary 3.1 (parity: forced real existence)**.** *The seven-point Hurwitz fiber H has a real point, independently of the explicit computation of κ.*

Proof. Complex conjugation is an involution, so its orbits have sizes 1 or 2. Hence $\#H(\mathbb{R}) + 2m = 7$ for some integer $m \geq 0$; the number of fixed points is odd and positive. This uses only the seven-point census and stability under complex conjugation. □

Theorem 3.2. *On the seven classes of Proposition 2.2, complex conjugation acts by the involution $\kappa = (1\,2)(5\,7)$, fixing the labels 3, 4, 6.*

Proof (certificate). On the anchored slice the map takes the pleasant form $y \mapsto g_1 y$: indeed $\kappa(g_1, y, z) = (g_1, z^{-1}, y^{-1})$ with $z^{-1} = (g_1 y)^{-1}\,{}^{-1} = g_1 y$. For each label i an explicit $c_i \in C$ with

$$c_i^{-1}\,(g_1 y_i)\,c_i \;=\; y_{\sigma(i)}, \qquad \sigma = (1\,2)(5\,7),$$

is exhibited in the artifact repository and re-verified by the Magma referee; by Proposition 2.2(1) this determines the action on classes. Involutivity of the induced σ and the membership certificates $c_i \in C \cap M_{23}$ are part of the same receipt. □

Remark 3.3 (forced, computed, arithmetic)**.** Corollary 3.1 forces $\#H(\mathbb{R}) \geq 1$, whereas Theorem 3.2 computes $\#H(\mathbb{R}) = 3$. Equivalently, the degree-7 étale coordinate algebra of H has exactly three real embeddings, counted across its residue fields (fields of moduli). A $\mathbb{Q}$-point requires a geometric point fixed by every element of $G_{\mathbb{Q}}$, rather than by complex conjugation alone. Thus the exact certificates restrict possible rational points to labels $3, 4, 6$, but do not imply that any of them is rational. They do not exclude a transitive seven-point orbit, whose residue field would have signature $(3, 2)$ (Corollary 3.4).

Label 6 is the $G_{\mathbb{Q}}$-fixed point established independently by [HJLPPZ]; that $\kappa(6) = 6$ is a consistency check the certificate passes, not an input. Its uniqueness and the remaining sextic orbit are the stronger arithmetic conclusions of Numerical Result 6.1, subject to the stated integer-relation recognitions.

Corollary 3.4 (partition menu)**.** *Every $G_{\mathbb{Q}}$-orbit is a union of κ-blocks $\{1, 2\}$, $\{5, 7\}$, $\{3\}$, $\{4\}$, $\{6\}$.*

(1) *From Theorem 3.2 alone, the partition shape of the seven is one of the twelve shapes* 2+2+1+1+1, 2+2+2+1, 3+2+2, 3+2+1+1, 3+3+1, 4+1+1+1, 4+2+1, 4+3, 5+1+1, 5+2, 6+1, 7.

(2) *Since label 6 is $G_{\mathbb{Q}}$-fixed* [HJLPPZ]*, the orbit $\{6\}$ splits off and the partition of the remaining six is one of the eight shapes*

$$2{+}2{+}1{+}1, \quad 2{+}2{+}2, \quad 3{+}2{+}1, \quad 3{+}3, \quad 4{+}1{+}1, \quad 4{+}2, \quad 5{+}1, \quad 6.$$

Numerical Result 6.1 selects the true shape.

[1] The generic form of the formula is (x^{-1}, z^{-1}, y^{-1}); here x is an involution.

4. Power series models of the seven quotient curves

Let $\Delta = \Delta(23, 2, 23)$ be the hyperbolic triangle group with the order-23 vertex uniformizing the branch point $\sqrt{-23}$ (slot $23A$), and for each label i let $\Gamma_i \leq \Delta$ be the index-23 subgroup determined by the permutation triple of (g_1, y_i). The quotient $X_i = \Gamma_i \backslash \mathcal{H}$ is the genus-4 quotient curve of the i-th cover; by Riemann–Hurwitz $g = 1 + \frac{1}{2}(23 - 1 - 15 - 1) = 4$, where 15 is the number of cycles of the $2A$-permutation.

We compute a basis of the weight-2 cusp forms $S_2(\Gamma_i) = H^0(X_i, \Omega^1)$ as power series in the disc uniformizer w centered at the elliptic fixed point of the order-23 vertex, following [KMSV14, VW14]: with $\rho = 0.9812834897968066\ldots$ (the radius of the fundamental domain), truncation order $N = 12187$, and $Q = 12207$ collocation points on $|w| = \rho$, the coefficient vector of each form is a fixed point of a structured evaluation–interpolation operator A, computed by the Arnoldi iteration implemented in the standalone solver of [Bel] at working precision 100 digits.

Validation is layered:

(1) the geometry stack (coset representatives, fundamental-domain reduction, automorphy factors) passes a battery of exact receipts, including the identity $\mathrm{Im}(\gamma z) = \mathrm{Im}(z)/|cz + d|^2$ at every collocation point and the radius bound $|w_{\mathrm{red}}| \leq \rho$;
(2) at low precision the operator A, rebuilt from the emitted input file alone, has numerical fixed space of dimension exactly $4 = \dim S_2(\Gamma_i)$, with singular-value separation 10^{-15} vs. $10^{-1.1}$ — an end-to-end structural check;
(3) the 100-digit solve validates each returned vector x by the finite-operator residual $\|Ax - x\|/\|x\|$; for label 6 all four residuals are $\leq 10^{-91}$, against an acceptance gate of 10^{-88} (this validates the numerical operator, not the truncation-to-algebraic step, which is addressed in §5);
(4) the 100-digit coefficients agree with the independently computed low-precision fixed space to $\sim 10^{-13}$, the roundoff floor of the latter.

For the remaining labels the validated residuals (item 3) are 1.2×10^{-91} (label 1), 4.6×10^{-92} (2), 3.3×10^{-92} (3), 8.9×10^{-92} (4), 1.7×10^{-92} (5), and 2.3×10^{-92} (7), all against the same 10^{-88} gate.

5. Marked Petri invariants

Write f_0, f_1, f_2, f_3 for the basis of $S_2(\Gamma_i)$ echelonized with respect to the w-expansion, $f_j = w^j + O(w^4)$. Since X_i is not hyperelliptic (this is checked, not assumed: see below), the canonical embedding $X_i \hookrightarrow \mathbb{P}^3$ realizes X_i as the complete intersection of a unique quadric Q_i and a cubic modulo Q_i (Petri). We compute $Q_i = \sum_{j \leq k} q_{jk}\, x_j x_k$ from the one-dimensional kernel of the evaluation matrix of the ten products $f_j f_k$ — the kernel dimension being exactly 1 is the non-hyperellipticity gate.

Lemma 5.1 (forced filtration structure)**.** *For every label, $q_{00} = q_{01} = 0$, $q_{02} = -q_{11}$, and $q_{03} = -q_{12}$ (in the echelon basis, after scaling).*

Proof. The relation $\sum q_{jk} f_j f_k = 0$ has w-expansion beginning

$$q_{00} + q_{01} w + (q_{02} + q_{11}) w^2 + (q_{03} + q_{12}) w^3 + O(w^4),$$

because $f_j f_k = w^{j+k} + O(w^4)$ and no products contribute below order 4 otherwise. Each displayed coefficient must vanish. □

Numerically the four vanishing combinations are observed at 10^{-92}, an additional receipt. The moduli-carrying data is thus the vector $(q_{11}, q_{12}, q_{13}, q_{22}, q_{23}, q_{33})$ up to the following normalization freedom.

Lemma 5.2 (weights)**.** *The echelon basis is unique given the local parameter w; the residual freedom is $w \mapsto \lambda w$ ($\lambda \in \mathbb{C}^*$), under which the re-echelonized basis satisfies $f_j \rightsquigarrow \lambda^{-j} f_j(\lambda\,\cdot)$ and the quadric coefficients transform by*

$$q_{jk} \;\longmapsto\; \mu\,\lambda^{j+k}\,q_{jk} \qquad (\mu \in \mathbb{C}^* \text{ a common scale}).$$

Consequently any ratio $\prod q_{j_t k_t} \big/ \prod q_{l_t m_t}$ with equally many factors up and down and matching total weight $\sum(j_t + k_t) = \sum(l_t + m_t)$ is independent of all normalization choices. (The sign of the weight exponent depends on whether one transforms functions or differentials; only weight differences *matter for such ratios — see Convention 5.5 for the form-level convention used in the algebraic argument.)*

We call such ratios *marked invariants*. The next proposition — the hinge of the method — shows they are algebraic and G_K-equivariant. Its proof is purely algebraic: no numerical statement enters. It rests on three observations: adapted parameters form a μ_{23}-torsor; the analytic parameter agrees with any adapted one to order > 12; and on a genus-4 curve the Petri quadric is determined by 12-jets, since $\deg 2K_X = 12$. The structural reason the argument closes is the inequality

$$\boxed{23 \;>\; \deg 2K_X \;=\; 4g-4 \;=\; 12\,.}$$

Definition 5.3 (adapted parameter)**.** Let (X, ϕ) be a cover in the family over a finite extension L/K contained in $\overline{\mathbb{Q}}$, and P the unique (totally ramified) point above $\sqrt{-23}$. A formal parameter t at P is *adapted* if $\phi = \sqrt{-23} + t^{23}$ exactly in $\widehat{\mathcal{O}}_{X_{\overline{L}},P}$. Write $\mathscr{T}_P = \{t \text{ adapted}\}$.

Lemma 5.4 (torsor)**.** *The functor $\mathscr{T}_P$ is a μ_{23}-torsor over L — any two adapted parameters differ by $t \mapsto \mu t$, $\mu^{23} = 1$ — and $\mathscr{T}_P(\overline{L}) \neq \varnothing$. Equivalently: adapted parameters exist after a finite extension of L, and need not exist over L itself. The torsor is functorial under $\mathrm{Gal}(\overline{L}/L)$ for $L \supseteq K$: if t is adapted for (X, ϕ, P) then t^σ is adapted for $(X^\sigma, \phi^\sigma, P^\sigma)$ (using $\sigma(\sqrt{-23}) = \sqrt{-23}$). Moreover, if w is the analytic disc parameter of §4 (canonical up to $w \mapsto \lambda w$), then every adapted t satisfies*

$$t \;=\; \lambda\, w\,\big(1 + u_1 w^{23} + u_2 w^{46} + \cdots\big), \qquad dt \;=\; \lambda\,\big(1 + O(w^{23})\big)\,dw$$

for some $\lambda \in \mathbb{C}^*$: the comparison unit is a power series in w^{23}.

Proof. Writing $\phi - \sqrt{-23} = c\,\tilde{t}^{\,23}\big(1 + h(\tilde{t})\big)$ in any parameter $\tilde{t}$ over L, an adapted parameter is $t = \tilde{t}\,\big(c(1+h)\big)^{1/23}$; the 23rd root of the unit $1 + h$ exists in $L[[\tilde{t}\,]]$ (characteristic 0), while the root $c^{1/23}$ in general requires a finite extension — whence existence over $\overline{L}$ only, and the μ_{23}-torsor structure. Functoriality is immediate from $\sigma(\phi - \sqrt{-23}) = \phi^\sigma - \sqrt{-23}$. For the analytic comparison: $\phi - \sqrt{-23}$ is invariant under the order-23 rotation $w \mapsto \zeta_{23} w$ about the disc center, hence is a power series in w^{23}; extracting the 23rd root as above gives the displayed form, and dt follows by differentiation. □

Convention 5.5 (weights on coefficients)**.** Fix a strictly increasing vanishing sequence $0 \leq a_0 < a_1 < a_2 < a_3$ (the orders of vanishing at P realized on $H^0(X, \Omega^1)$; generically, and for all our labels by a per-label numerical gate, $a_j = j$). The echelon basis w.r.t. a parameter t

is $\omega_j = \big(t^{a_j} + \sum_{n>a_j} a_{j,n} t^n\big)dt$ with $a_{j,a_k} = 0$ for $k \neq j$. Under $t \mapsto \mu t$, after renormalizing leading terms,

$$a_{j,n} \;\longmapsto\; \mu^{\,a_j - n}\, a_{j,n},$$

and the Petri coefficients transform by $q_{jk} \mapsto \tilde{\mu}\,\mu^{-(a_j+a_k)} q_{jk}$ for a common scalar $\tilde{\mu}$; only weight *differences* matter below, so the sign convention in the exponent is immaterial for weight-matched ratios. A ratio $\prod q_{j_t k_t} / \prod q_{l_t m_t}$ is *weight*-0 if the factor counts agree and $\sum(a_{j_t} + a_{k_t}) = \sum(a_{l_t} + a_{m_t})$.

Lemma 5.6 (Petri from 12-jets)**.** *Let X be a non-hyperelliptic curve of genus 4 and $P \in X$ any point. The jet map $j^{12}_P \colon H^0(X, 2K_X) \to (2K_X) \otimes \mathcal{O}_{X,P}/\mathfrak{m}_P^{13}$ is injective, and consequently the Petri line $\ker\big(m \colon \operatorname{Sym}^2 H^0(K_X) \to H^0(2K_X)\big)$ equals $\ker\big(j^{12}_P \circ m\big)$: the quadric is determined exactly by the coefficients of the products $\omega_j\omega_k$ through order 12 at P.*

Proof. $\deg 2K_X = 4g - 4 = 12$, so a nonzero global section of $2K_X$ vanishes at P to order at most 12; a section with vanishing 12-jet is therefore zero. The second statement is immediate. □

Proposition 5.7 (equivariance)**.** *Let J be any weight-0 ratio in the sense of Convention 5.5, defined on the locus where its denominator factors are nonzero. Then:*

1. *$J(X, \phi) \in \overline{\mathbb{Q}}$ is an invariant of the isomorphism class of the marked cover (X, ϕ, P), and for every $\sigma \in G_K$ the square*

$$\begin{array}{ccc} (X, \phi, P) & \xmapsto{\;J\;} & \overline{\mathbb{Q}} \\ \big\downarrow{\scriptstyle \sigma} & & \big\downarrow{\scriptstyle \sigma} \\ (X^\sigma, \phi^\sigma, P^\sigma) & \xmapsto{\;J\;} & \overline{\mathbb{Q}} \end{array}$$

   commutes;
2. *the analytically computed value of §4 equals $J(X, \phi)$.*

Proof. (1) Pull back to the torsor: for an adapted $t \in \mathscr{T}_P$ (over $\overline{L}$, Lemma 5.4) the flag of vanishing orders, the echelon basis, the products $\omega_j\omega_k \in H^0(2K_X)$, their 12-jets at P in the parameter t, and the Petri line recovered from those jets (Lemma 5.6) are all algebraic constructions, compatible with any field embedding; hence all $a_{j,n}$ and q_{jk} lie in $\overline{\mathbb{Q}}$ and depend on the choice of t only through the μ_{23}-action of Convention 5.5. A weight-0 ratio is μ_{23}-invariant, so J descends from the torsor to the marked cover itself. Functoriality of $\mathscr{T}_P$ under G_K (Lemma 5.4) then gives commutativity: σ carries the entire diagram

$$\begin{array}{ccccc} \operatorname{Sym}^2 H^0(K_X) & \xrightarrow{\;m\;} & H^0(2K_X) & \xrightarrow{\;j^{12}_P\;} & J^{12}_P(2K_X) \\ \big\downarrow{\scriptstyle \sigma} & & \big\downarrow{\scriptstyle \sigma} & & \big\downarrow{\scriptstyle \sigma} \\ \operatorname{Sym}^2 H^0(K_{X^\sigma}) & \xrightarrow{\;m^\sigma\;} & H^0(2K_{X^\sigma}) & \xrightarrow{\;j^{12}_{P^\sigma}\;} & J^{12}_{P^\sigma}(2K_{X^\sigma}) \end{array}$$

to the corresponding diagram for (X^σ, ϕ^σ), with the right-hand horizontal maps injective, and the Petri line functorially recovered on each row.

(2) By Lemma 5.4, $t = \lambda w(1 + O(w^{23}))$ and $dt = \lambda(1 + O(w^{23}))dw$, so the transition between the echelon computations in w and in t differs from the diagonal scaling $\operatorname{diag}(\lambda^{a_j})$ only at jet orders ≥ 23. Since $23 > 12$, the two computations have *identical* 12-jets of the products $\omega_j\omega_k$ up to that diagonal scaling and a common scalar; by Lemma 5.6 the Petri lines agree up to the same scaling, and weight-0 ratios kill it. No truncation hypothesis is

needed: the entire $2K$-linear system is blind to the nonlinear discrepancy between w and t, precisely because $23 > \deg 2K_X$. □

Remark 5.8 (numerical cross-check)**.** Although the proof above is exact, the pipeline still runs the corresponding receipt: for label 6 the quadric recomputed from coefficient orders ≤ 22 alone agrees with the full computation to 3×10^{-92} and the invariants to 7×10^{-91}, consistent with (and now implied by) Lemma 5.6.

Remark 5.9 (the field is K, not $\mathbb{Q}$)**.** Even for the $G_{\mathbb{Q}}$-fixed label 6 the marked invariants need not be rational: the marked triple (X, ϕ, P) is a K-object, since P lies over $\sqrt{-23}$. The computation below finds them in K exactly. Complex conjugation ($\sigma \notin G_K$) sends the $\sqrt{-23}$-marked data of label i to the $-\sqrt{-23}$-marked data of label $\kappa(i)$; the resulting compatibility between labels $\{1,2\}$ and $\{5,7\}$ is a numerical gate in §6, not an assumption.

We use the six weight-0 ratios

$$J_1 = \frac{q_{13}}{q_{22}}, \quad J_2 = \frac{q_{11}q_{13}}{q_{12}^2}, \quad J_3 = \frac{q_{11}q_{33}}{q_{13}q_{22}}, \quad J_4 = \frac{q_{12}q_{23}}{q_{13}q_{22}}, \quad J_5 = \frac{q_{11}q_{23}}{q_{12}q_{13}}, \quad J_6 = \frac{q_{22}^2}{q_{11}q_{33}}.$$

A numerical gate checks, for every label, that each denominator factor appearing in $J_1, \ldots, J_6$ is bounded away from zero (receipts), so the six ratios are defined at all seven points.

5.1. **Ground-truth gate: label 6.** For label 6, PSLQ at 100 working digits recognizes all six invariants in K with small height:

invariant	value in $K = \mathbb{Q}(\sqrt{-23})$
$J1$	$\dfrac{-4152 + 693\sqrt{-23}}{10237}$
$J2$	$\dfrac{-148227 - 34830\sqrt{-23}}{142129}$
$J3$	$\dfrac{-32137795 - 2206259\sqrt{-23}}{18856554}$
$J4$	$\dfrac{-44137861 - 681405\sqrt{-23}}{18856554}$
$J5$	$\dfrac{113993 - 1027623\sqrt{-23}}{694434}$
$J6$	$\dfrac{183063721 + 14235727\sqrt{-23}}{182131716}$

Each recognition leaves ~ 85 digits of confirmation margin. As a negative control, a ratio with mismatched weight (hence still containing the transcendental λ) fails recognition, exactly as Lemma 5.2 predicts.

6. The Galois orbit partition

For each invariant J the multiset $\{J(X_1), \ldots, J(X_7)\}$ (all marked at the $\sqrt{-23}$-side) is G_K-stable, so

$$\Phi_J(T) \;=\; \prod_{i=1}^{7} \big(T - J(X_i)\big) \;\in\; K[T],$$

and the degrees of the irreducible factors of Φ_J over K compute the G_K-orbit partition of the seven marked covers (provided J separates the seven, which is checked numerically). The $G_{\mathbb{Q}}$-partition then follows by gluing with Theorem 3.2: $G_{\mathbb{Q}}$-orbits are the κ-stable unions of G_K-orbits generated by the pairs $\{1,2\}$, $\{5,7\}$.

Gates: (i) agreement of the partition across the independent invariants $J_1, \ldots, J_6$; (ii) the κ-consistency check that the directly-computed invariants of labels 2 and 7 are the appropriate Galois conjugates of those of 1 and 5; (iii) a precision ladder (recognition at 60 digits, confirmation at 100; escalation to 200 digits if coefficient heights require it).

From partition to permutation group. The factor degrees of Φ_J give only the orbit *sizes*. From the recognized (and consistency-checked) factors we extract more: the Galois group of each irreducible factor over K, discriminants, and Frobenius cycle types at a sample of good primes of K determine the image

$$A_K \;=\; (\text{the permutation group induced by } G_K \text{ on the seven}) \;\leq\; S_7$$

up to conjugacy, and adjoining the unconditionally known complex-conjugation permutation gives $A_{\mathbb{Q}} = \langle A_K, \kappa \rangle = \rho(G_{\mathbb{Q}})$ as an *explicit subgroup of* S_7. It is this group, not merely the partition shape, that is to be placed alongside the ambient bound of §7.

Epistemic status and the exactification path. We state the outcome as a *high-precision numerical determination*: by Proposition 5.7 the quantities $J(X_i)$ are honest algebraic numbers and the set-level Galois structure is as described, but their *identification* with specific elements of $\overline{\mathbb{Q}}$ rests on integer-relation detection (PSLQ margin plus cross-implementation agreement). This is compelling evidence, not yet proof. Two upgrades convert it to a theorem. *Cheap consistency*: the recognized values must satisfy exact algebraic identities — the recognized Φ_J must have K-coefficients matching the recognized roots symmetrically, and distinct invariants satisfy resultant compatibilities — all checkable in exact arithmetic; failures would falsify recognition. *Definitive*: reconstruct from the recognized invariants exact models (Q_i, C_i) over the recognized fields by lattice reduction, and verify *algebraically* that each defines a genus-4 curve whose associated cover has monodromy M_{23} of type $(2A, 23A, 23B)$ — at which point the partition statement inherits a full proof. The first upgrade is carried out here: the recognized sextic coefficients are the exact symmetric functions of the six recognized roots by construction, and the two independently recognized sextics satisfy an exact compatibility, verified purely symbolically — f_2 acquires a linear factor over $\mathbb{Q}[x]/(f_1)$, so the two independent descended coordinates generate the *same* sextic field F (receipt `partition-p100-receipt.json`). The second upgrade is future work in the spirit of [HJLPPZ]'s own exactness step.

Descent to $\mathbb{Q}$: trace and norm along the marking. Let A denote the degree-7 étale $\mathbb{Q}$-algebra of the unmarked Hurwitz fiber; the $\sqrt{-23}$-marked fiber has coordinate algebra $A_K = A \otimes_{\mathbb{Q}} K$ with $K/\mathbb{Q}$-involution τ, and Proposition 5.7 realizes each marked invariant as an element $J \in A_K$. The elements

$$S = J + \tau J, \qquad N = J \cdot \tau J$$

are the quadratic trace and norm and lie in A: they are $G_{\mathbb{Q}}$-equivariant functions of the *unmarked* points. Pointwise, by Theorem 3.2, τ acts by $(\tau J)(i) = \overline{J(\kappa(i))}$, so S and N are computable from the $\sqrt{-23}$-marked data alone. (The assembled polynomials $\prod_i (T - S(i))$ have real coefficients *by construction* — receipt `DESCENT_ASSEMBLY_REALITY`, an implementation gate, not an equivariance test.)

The result. Among small integer combinations of the twelve descended coordinates S_r, N_r $(r = 1, \ldots, 6)$, the coordinate $U_1 = S_1 + N_1$ (and, independently, $U_2 = S_2 + N_2$) separates the seven points and produces exact recognitions with ~ 70 digits of margin at working precision 100:

Numerical Result 6.1 (high-precision determination of the orbit partition)**.** *Subject to the integer-relation recognitions below (margins* 10^{-88}–10^{-90}*, all gates passed), the* $G_{\mathbb{Q}}$*-set of the seven Hurwitz points is*

$$\{6\} \sqcup \{1,2,3,4,5,7\},$$

i.e. the partition is $1+6$*: label* 6 *is the unique rational point and the remaining six form a single* $G_{\mathbb{Q}}$*-orbit. Moreover the associated permutation images are*

$$\rho(G_{\mathbb{Q}}) \;=\; \mathrm{Stab}_{S_7}(6) \;\cong\; S_6, \qquad \rho(G_K) \;=\; S_6 \text{ as well.}$$

Equivalently, $H \simeq \operatorname{Spec}\mathbb{Q} \sqcup \operatorname{Spec} F$ *with* $F = \mathbb{Q}[T]/(f)$ *sextic,* $\mathrm{Gal}(f^{\mathrm{spl}}/\mathbb{Q}) = S_6$*, and signature* $\mathrm{sig}(F) = (2,2)$ *(forced by Theorem 3.2).*

For the remainder of §§6–7, assertions identifying F with the residue field of the six-point Hurwitz orbit are conditional on the integer-relation recognitions of Numerical Result 6.1; statements about the displayed polynomials f, f_2 themselves are exact.
Evidence and gates:

(1) $U_1(6) = -\frac{5541}{10237}$ exactly (equal to $S_1(6) + N_1(6)$ from the label-6 ground truth of §5), and the six values $U_1(i)$, $i \neq 6$, are the roots of the exact monic sextic with common denominator 50019668646075143:

$f(T) = T^6 + \frac{1}{50019668646075143}\sum_k c_k T^k$ with c_k:	
T^0	−139453053861313073
T^1	−868575878997768238
T^2	−759255603013669319
T^3	−55027953306246948
T^4	353749018385880145
T^5	206305139624557234

Each of the six independently computed roots satisfies f to 6×10^{-88}.

(2) f is irreducible over $\mathbb{Q}$ with Gal $= S_6$ (6T16), verified independently by SymPy, by a Frobenius inventory (below), and by a Magma referee run (receipt `magma-sextic-referee-ledger.txt`).

(3) Replication and field identity: $U_2 = S_2 + N_2$ independently yields a second exact irreducible sextic f_2 with Galois closure of group S_6 and $U_2(6) = \frac{54447}{142129}$, consistent with the label-6 anchors. Moreover — purely symbolically, with no numerical input — f_2 acquires a linear factor over $\mathbb{Q}[x]/(f_1)$: the two independently descended coordinates generate the *same* sextic field F. This compatibility is an exact statement about the two displayed polynomials, independent of any numerics; the explicit isomorphism certificate is displayed in Appendix C.

(4) Frobenius inventory over 199 good primes (the first 199 primes coprime to the sextic denominator and to the discriminant numerator, excluding 23): nine distinct cycle types observed, with frequencies consistent with S_6 Chebotarev densities (the two absent classes are those of smallest density). Over the 94 *split* primes $\left(\frac{-23}{p}\right) = +1$, whose Frobenii sample $\rho(G_K)$, the *odd* classes $(1,2,3)$, (6), $(1,1,4)$ occur 52 times. This inventory serves as replication; the exact statement is Proposition 6.2 below.

(5) The commutator-class experiment of Remark 7.3 is settled: the six-point orbit crosses the partition $\{1,2,3,4\} \mid \{5,6,7\}$, so the arithmetic action does not stabilize it.

Proposition 6.2 (exact form of $\rho(G_K) = S_6$)**.** *Assume the recognitions of Numerical Result 6.1, and let* L *be the splitting field of* f*, so* $\mathrm{Gal}(L/\mathbb{Q}) = S_6$*. Then* $L \cap K = \mathbb{Q}$ *and* $\mathrm{Gal}(LK/K) \cong S_6$*; consequently* $\rho(G_K) = S_6$*.*

Proof. An S_6-extension of $\mathbb{Q}$ has a unique quadratic subfield, the fixed field of A_6, namely $\mathbb{Q}(\sqrt{\operatorname{disc} f})$. Complex conjugation acts on the six roots with cycle type $2^2 1^2$ (Theorem 3.2), an *even* permutation; it therefore lies in A_6 and fixes the quadratic subfield pointwise, so that subfield is real (equivalently $\operatorname{disc} f > 0$, as the signature $(2,2)$ forces). Since K is imaginary quadratic, $K \not\subset L$, hence $L \cap K = \mathbb{Q}$ and restriction is an isomorphism $\operatorname{Gal}(LK/K) \xrightarrow{\sim} \operatorname{Gal}(L/\mathbb{Q}) = S_6$. The action of G_K on the six points is the action of $\operatorname{Gal}(LK/K)$ on the roots of f, which is therefore all of S_6. □

Remark 6.3 (the compositum explains the degree-12 wall)**.** F has two real embeddings, hence contains no imaginary quadratic field; so $F \cap K = \mathbb{Q}$ and $[FK : \mathbb{Q}] = 12$, $[FK : K] = 6$. The $\sqrt{-23}$-marked invariants of the six non-rational points naturally generate subfields of FK: the degree-12 heights first encountered by the marked computation were the compositum announcing itself.

Local structure at 23. The rescaling $t \mapsto t/\sqrt{-23}$, defined over K, moves the branch locus to $(0, 1, -1)$: the apparent collision of $0, \pm\sqrt{-23}$ modulo the prime above 23 is a coordinate artifact, not an intrinsic degeneration of the configuration. (No good-reduction claim at other primes is intended: in the rescaled model 1 and -1 collide modulo 2, and 2 divides the order of the $2A$ monodromy; no prime other than 23 is analyzed here.) What singles out 23 is that it divides *two* of the three local monodromy orders — the setting of stable reduction of three-point covers with cyclic p-Sylow subgroup (Raynaud; Bouw–Wewers; Obus [Ob11a, Ob11b]). At $p = 23$ the local decomposition of the Hurwitz algebra is strikingly asymmetric.

Proposition 6.4 (local factorization at 23)**.** *$F \otimes \mathbb{Q}_{23} \simeq F_2 \times F_4$ with $(e, f) = (2, 1)$ and $(4, 1)$ respectively; both extensions are totally ramified and tame. Hence the seven-point Hurwitz algebra decomposes locally as*

$$\mathbb{Q}_{23} \times F_2 \times F_4,$$

tame inertia at 23 *acts on the seven points with cycle type* $(1)(2)(4)$*, and* label 6 is the unique inertia-fixed point of the fiber at 23.

Proof (certificate). $f \bmod 23 = (x + 13)^2(x + 9)^4$. The quartic cluster has a single Newton segment of slope $-1/4$ (Eisenstein–Dumas: a totally ramified quartic); the quadratic cluster has odd 23-adic discriminant valuation ($v_{23}(\operatorname{disc} f) = 6 = 3+3$ with vanishing cross-resultant valuation, the clusters lying at distinct residues), hence is ramified. Both indices are prime to 23 (tame). The factorization, ramification indices, and residue degrees are independently confirmed by a Magma computation over $\mathbb{Q}_{23}$ (`magma-local23-referee-ledger.txt`). □

Remark 6.5 (towards a mechanism)**.** We emphasize the careful reading: Proposition 6.4 exhibits label 6 as the unique *inertia*-fixed point at 23 — a local distinction, weaker than global rationality, but the first structural property found that singles out label 6. Two remarks situate it in the stable-reduction theory for cyclic p-Sylow (here $P = C_{23}$ of order exactly 23, $N_{M_{23}}(P) \cong 23{:}11$, so $m = |N/C| = 11$). First, the proven theorems of [Ob11a, Ob11b] do not cover M_{23}: part I assumes G p-solvable, part II assumes $m = 2$. However, for $|P| = p$ their conclusion pattern — vanishing of the first higher ramification groups, i.e. *tameness* of the field of moduli at p — is exactly what Proposition 6.4 establishes for M_{23} with $m = 11$: our data is a new data point consistent with the expected phenomenology outside the proven range. Second, the sharply posed open question: *derive the inertia cycle type* $(1)(2)(4)$ — *and the identity of the fixed point — from the deformation data of the stable model at* 23*, without using* f. An affirmative answer would explain why this point is locally distinguished; the remaining gap to global rationality would then be the passage from the decomposition group at 23 to all of $G_{\mathbb{Q}}$.

7. Finite Grothendieck–Teichmüller constraints

The computations above determine *what* the Galois set of seven points is; the question of [HJLPPZ] — *why* is there a fixed point — asks for a structural cause. In this section we prove a sharp negative result: the finite-level Grothendieck–Teichmüller constraints attached to M_{23} impose *no* restriction at all on permutations of the seven. Whatever forces the fixed point is invisible at this level.

For a finite group G, Guillot [Gui] defines groups $\mathcal{GT}(G) \supseteq \mathcal{GT}_1(G)$ acting on the set $\mathscr{P}_c(G)$ of generating pairs of G up to simultaneous conjugacy, compatibly with the tower defining Drinfeld's $\widehat{\mathcal{GT}} \supseteq G_{\mathbb{Q}}$; the operators $\theta \cdot [g,h]_c = [h,g]_c$ and $\delta \cdot [g,h]_c = [h^{-1}g^{-1}, h]_c$ generate an S_3 of tautological symmetries. For G simple and non-abelian he proves the remarkable identification

$$\mathcal{GT}_1(G) \;\cong\; \mathscr{S}(G) \;:=\; C_{\mathrm{Sym}(\mathscr{P}_c)}\big(\mathrm{Out}(G) \times \langle \theta, \delta \rangle\big) \;\cap\; Y,$$

where Y is the subgroup preserving the partition of $\mathscr{P}_c$ by ordered class-vectors, and moreover [Gui, Prop. 4.12] that $\mathscr{S}(G)$ is a product of wreath products $E_k \wr S_{r_k}$, one factor per equivalence class of $\langle \theta, \delta \rangle \times \mathrm{Out}(G)$-orbits, with E_k a subquotient of the acting group.

Theorem 7.1 (no constraint at the M_{23}-level)**.** *The image of $\mathcal{GT}_1(M_{23})$ in the symmetric group of the set $\mathrm{Ni}(2A, 23A, 23B)^{\mathrm{in}}$ is the full S_7.*

Proof. $\mathrm{Out}(M_{23}) = 1$, so the acting group is $H = \langle \theta, \delta \rangle \cong S_3$. The six ordered class-vectors in the H-orbit of $(2A, 23A)$ are pairwise distinct (inversion exchanges $23A$ and $23B$ since -1 is a quadratic non-residue mod 23), so H acts freely on the 42 pair-classes lying over them, with seven orbits — one for each Nielsen class — and every orbit meets each of the six class-vector blocks in exactly one point. These seven H-orbits are therefore equivalent in the sense of [Gui, Prop. 4.12], and the corresponding wreath factor of $\mathscr{S}(M_{23})$ is $E \wr S_7$, where E consists of the H-equivariant self-maps of one free orbit preserving each block singleton. An H-equivariant self-map of a free orbit is right translation by some $s \in H$, which preserves the six singletons iff $s = 1$; hence $E = 1$ and the factor is S_7, acting by arbitrary permutation of the seven orbits. Since $\mathcal{GT}_1(M_{23}) \cong \mathscr{S}(M_{23})$, the restriction to the seven Nielsen classes is all of S_7. □

Scope. Theorem 7.1 concerns the *single finite level* $\mathcal{GT}_1(M_{23})$. The comparison with the global Grothendieck–Teichmüller action is inverse-limit in nature: information at different finite levels must satisfy simultaneous compatibility conditions, and cross-level compatibilities (for instance those exploited by Belyi-extending constructions, which compare the M_{23}-projection with projections at other monodromy groups) are *not* eliminated by Theorem 7.1. What the theorem does show is that every invariant internal to the single level — every quantity attached functorially to $\mathscr{P}_c(M_{23})$ alone — is constant on the seven. Our finite-depth searches through cross-level constructions (postcomposition by Belyi-extending maps of outer degree ≤ 12 with all recolorings; square-root and pullback probes) found no constraint either, but that remains an empirical statement about finite search depth, not a structural one.

Remark 7.2 (cyclotomic remark, in its correct strength)**.** $K = \mathbb{Q}(\sqrt{-23})$ is the quadratic subfield of $\mathbb{Q}(\zeta_{23})$, and $23A$, $23B$ are exchanged exactly by non-residue powering. For $\sigma \in G_K$ the mod-23 cyclotomic character lies in $(\mathbb{F}_{23}^{\times})^2 \cong C_{11}$, and since $11 \nmid 7!$, *no nontrivial permutation of the seven can factor solely through the quadratic-residue cyclotomic quotient.* This does not decompose a Galois element into a cyclotomic part and an f_σ-part — the permutation induced by (λ, f) depends on both jointly — and we do not claim the "cyclotomic part" of the action is trivial in any stronger sense.

Remark 7.3 (a separating, non-GT invariant)**.** The class of the commutator $[x, y]$ *does* separate the seven: labels $1, 2, 3, 4$ have $[x, y]$ of order 4 (cycle type $4^4 2^2 1^3$) and labels $5, 6, 7$ of order 6 (type $6^2 3^2 2^2 1$), uniformly over all six recolorings. By Theorem 7.1 this partition *cannot* be $\mathcal{GT}_1(M_{23})$-invariant — an instructive example of how natural generating-pair invariants fail to be Galois-functorial. The computation of §6 settles the question negatively: the six-point $G_{\mathbb{Q}}$-orbit crosses $\{1, 2, 3, 4\} \mid \{5, 6, 7\}$, so the arithmetic action does *not* stabilize this partition (it is merely κ-stable).

The sharpened question. The comparison with the inverse limit requires care about which Galois group enters: the tower of level-one groups $\mathcal{GT}_1(G)$ receives not $G_{\mathbb{Q}}$ itself but its derived subgroup $G'_{\mathbb{Q}}$ [Gui]. Since $\rho(G_{\mathbb{Q}}) \cong S_6$ surjectively onto the six-point orbit (Numerical Result 6.1), the image of the derived subgroup is $\rho(G'_{\mathbb{Q}}) = [S_6, S_6] = A_6$, and the correct chain is

$$A_6 \;=\; \rho(G'_{\mathbb{Q}}) \;\subseteq\; \mathrm{im}\big(\widehat{\mathcal{GT}}_1 \to \mathcal{GT}_1(M_{23})\big)\Big|_H \;\subseteq\; \mathcal{GT}_1(M_{23})\Big|_H \;=\; S_7,$$

in which Theorem 7.1 identifies the outermost group and eliminates it as an explanation: *the existence of the rational point is not forced by the M_{23}-level Grothendieck–Teichmüller constraints.* The single finite level places no restriction on the seven points. The image of the global inverse-limit group is potentially smaller — any of A_6, S_6, A_7, S_7 is compatible with the chain — but determining it requires cross-level compatibility information not computed here.

Numerical Result 6.1 determines the arithmetic image itself, and the answer is as sharp as it could be:

$$A_{\mathrm{arith}} \;=\; \rho(G_{\mathbb{Q}}) \;=\; \mathrm{Stab}_{S_7}(6) \;\cong\; S_6,$$

the largest subgroup of S_7 compatible with fixing label 6 — and already $\rho(G_K) = S_6$. The arithmetic action is maximally mixing everywhere the fixed point permits: the action on the six-point orbit is primitive, so it admits no nontrivial invariant partition, and the quadratic field K does not appear in the Galois closure of the sextic (Proposition 6.2). The question of [HJLPPZ] is thereby concentrated into a single sharp statement: *the entire symmetry breaking is*

$$S_7 \rightsquigarrow \mathrm{Stab}_{S_7}(6),$$

one rational point, and otherwise maximal Galois mixing. Why does arithmetic single out label 6 *— and nothing else?*

8. Verification and artifacts

All computations, receipts, and the Lean development are contained in the artifact repository `https://github.com/azide0x37/m23-seven-points`; releases of the repository are archived with a DOI. The receipt files are JSON with self-describing gates; each computational claim above names its receipt. Independent implementations: (a) a Python enumeration (bytes-encoded permutation BFS), (b) a from-scratch re-implementation used by the Lean certificate generator, (c) a Magma referee script (census, generation, and κ-certificates), run on the public Magma calculator. An in-progress Lean 4 development encodes and checks the small finite certificates (kernel-checked `decide` proofs, with a single quarantined `native_decide` enumeration); the full census build and the glue lemmas connecting those certificates to the paper's theorem statements are incomplete, and no end-to-end formal verification is claimed here.

The analytic layer is validated as described in §4; we emphasize that the final arbiter for every recognized algebraic number is integer-relation margin plus cross-implementation

agreement. A full replication of the pipeline at 200-digit working precision is in progress; its receipts will be added to the repository.

AI assistance and verification. This project made extensive use of AI systems, principally Claude (Anthropic) and ChatGPT (OpenAI), throughout the research process. Their roles included exploratory mathematical discussion, proposing and criticizing candidate approaches, literature triage, generation and debugging of computational code, orchestration and analysis of numerical experiments, and drafting and editing the manuscript under the author's direction. The author selected the research questions, determined which computational and mathematical claims to pursue, executed or supervised the computations, evaluated the resulting evidence, and takes responsibility for all claims in the paper. Computational claims are accompanied by reproducible artifacts and independent checks as described in this section; theoretical claims are supported by the proofs and references given in the text. AI-generated suggestions were not treated as mathematical evidence.

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Appendix A. Representatives and certificates

g_1 $(2A)$	`(1,11)(2,23)(3,8)(4,16)(5,21)(7,20)(15,19)(18,22)`
g_2 $(23A)$	`(1,2,11,10,16,9,6,3,23,19,20,14,21,17,4,8,22,5,18,15,13,7,12)`
g_3 $(23B)$	`(1,2,3,4,10,11,12,7,19,18,8,6,9,16,17,21,22,5,14,20,13,15,23)`

Anchored representatives (labels of Convention 2.1, with label 6 the class of the triple printed in [HJLPPZ]); every Nielsen class is represented by the pair (g_1, y_i):

label i	anchored y_i (so the pair is (g_1, y_i))
1	`(1,4,19,10,20,12,9,13,6,3,16,22,15,21,14,8,2,17,7,5,11,18,23)`
2	`(1,23,21,19,8,6,16,9,13,12,14,20,3,2,15,18,10,4,11,17,7,5,22)`
3	`(1,4,21,12,8,7,5,19,2,20,6,9,18,13,14,3,16,17,10,22,15,11,23)`
4	`(1,20,17,21,12,2,13,4,11,10,23,6,9,15,22,19,18,16,5,8,7,3,14)`
5	`(1,17,15,4,18,3,21,9,8,20,13,10,22,12,19,23,6,11,5,7,14,16,2)`
6	`(1,6,18,16,22,9,3,14,13,2,4,19,11,10,21,17,8,23,20,5,7,15,12)`
7	`(1,22,8,12,9,21,11,3,13,15,5,14,23,19,7,10,4,17,2,20,16,18,6)`

$C = \langle h_1, h_2 \rangle$, $h_1 =$ `(1,8,5,7)(2,23)(3,21,20,11)(4,15,16,19)(9,13)(10,14,12,17)`, $h_2 =$ `(1,4,3,22)(2,5,19,7)(8,18,11,16)(9,13)(14,17)(15,20,23,21)`.

κ-certificates ($\sigma = (1\,2)(5\,7)$):

i	c_i with $c_i^{-1}(g_1 y_i)c_i = y_{\sigma(i)}$
1	`(1,5,4,7,3,23)(2,11,21,16,20,8)(6,17,14)(9,12,13)(15,22)(18,19)`
2	`(1,23,3,7,4,5)(2,8,20,16,21,11)(6,14,17)(9,13,12)(15,22)(18,19)`
3	`(1,2)(3,8)(4,7)(6,17)(9,10)(11,23)(15,19)(16,20)`
4	`(1,5)(3,20)(4,16)(7,8)(10,12)(11,21)(14,17)(15,19)`
5	`(1,3,7,23,15,4)(2,19,16,11,8,20)(5,22)(6,9,10)(12,13,14)(18,21)`
6	`(1,11)(2,23)(4,7)(5,18)(6,10)(9,17)(16,20)(21,22)`
7	`(1,4,15,23,7,3)(2,20,8,11,16,19)(5,22)(6,10,9)(12,14,13)(18,21)`

Appendix B. Computation parameters

working precision	100 decimal digits
ρ (fundamental-domain radius)	$0.9812834897968066\ldots$
truncation order N	12187
collocation points Q	12207
residual gate	10^{-88}
label-6 residuals	$\le 10^{-91}$ (all four vectors)
low/high cross-check	9.9×10^{-14}

Appendix C. The field-identity certificate

Let α be a root of the sextic $f = f_1$ displayed in Numerical Result 6.1, and let f_2 be the second descended sextic:

$f_2(T) = T^6 + \frac{1}{1413856429158763966658103 61} \sum_k c_k T^k$ with c_k:	
T^0	41412828841714498436791938 1
T^1	$-$30148018897336133820545273694
T^2	82814356367377111157987163503
T^3	$-$10236272033398981465535697652
T^4	50609361904878698295312621 15
T^5	$-$48975052455725622785516462 2

The element $\beta = \sum_{k=0}^{5} r_k\, \alpha^k \in \mathbb{Q}[\alpha]/(f_1)$ with the coefficients below satisfies $f_2(\beta) = 0$. The verification is the exact polynomial-remainder identity $f_2(g(x)) \equiv 0 \pmod{f_1(x)}$ for $g(x) = \sum_k r_k x^k$ in $\mathbb{Q}[x]$ (receipt `iso-beta-receipt.json`); hence the two independently recognized sextics define the same field F.

k	r_k (numerator / denominator)
r_0	1139031854354334030139004760292567736326223637201380 98248180458436368958135756532517118245480750446 75 / 28760996192761906921309974353827872570823981550963 82 33246653433040914840393004142309629477814057615 36
r_1	-70055525742027620738029913063469887711965050852920 4 36382040680840049664033269663315666076259606117145 / 28760996192761906921309974353827872570823981550963 82 33246653433040914840393004142309629477814057615 36
r_2	-63138985976593313984233799422487340567671403983672 2 49772219471359832896209510086913254392530340893153 / 14380498096380953460654987176913936285411990775481 91 16623326716520457420196502071154814738907028807 68
r_3	12859312273628258937288329795515082234640996267374 86 85784929807295507478270787612979862862468269590 95 / 14380498096380953460654987176913936285411990775481 91 16623326716520457420196502071154814738907028807 68
r_4	42162305778518490776938441008880881279685365595005 13 28008606052650218446792632396105127902643583688 79 / 28760996192761906921309974353827872570823981550963 82 33246653433040914840393004142309629477814057615 36
r_5	27504286576311072628878696939201205855317719762485 52 39444109321284930781088274243147228225852595341 / 40508445341918178762408414582856158550456312043611 01 87671342863437908225905639637055816165935292416

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